TFG121-BF



Technology Overview

Terafence is proud to introduce a unique, secure, and innovative network security appliance – **Terafence TFG121-BF**, a dynamic, configurable, agile, and complete unidirectional Secure Gateway solution including the FPGA data-diode.

Terafence manufactures Cyber Security hardware products to allow security architects the freedom in protecting their networks as needed.

Terafence TFG121-BF is a stand-alone unit that offers unique features and capabilities as a unidirectional, data-diode based device. Acting as a full Protocol Breaker or proxy server, total network segmentation, Air-Gap level, is guaranteed.

TFG121-BF utilizes a proprietary FPGA-based hardware architecture, designed, and manufactured by **Terafence**.

Certifications & Compliances:

- **IEC62443-4-2 SL2**
- **FCC, EMI/EMC, CE, RoHS, MIL-STD-810F.**
- **Black Coated Aluminum chassis**

Supported Protocols

TFG121-BF supports multiple, simultaneous industrial and IT protocols, including:

- **SFTP/FTP/SMB –** Secure file or folder-tree transfers
- **SMTP –** Email relay
- **SYSLOG Forwarding –** Log event transmission
- **RTSP/ONVIF –** Live streaming for CCTV
- **HTTP/S –** File and data uploads to web servers or cloud platforms
- **Local SYSLOG Forwarding –** Internal log event management
- **Modbus –** Communication between multiple PLCs and HMIs
- **IEC-104 –** Protocol for power utilities
- **MQTT –** Data brokering and publishing
- And various other protocols as per customer requirements

Why choose TFG121-BF?

Key Features:

- Complete galvanic **network isolation** for enhanced security and protection.
- Handles **large file transfers** of 60 GB, 70 GB, or even larger files in a single instance.
- Complete **all-in-one solution** with **no** need for additional software or hardware.
- Proprietary **Terafence FPGA-based hardware** architecture for **hardware-enforced security.**
- Two dedicated CPUs for **protocol support, processing, and termination.**
- **Secure HTTPS-based web GUI** for configuration and management, accessible only from the secure side.
- **Hardened** operating system on the dedicated CPUs.
- Core security hardware operates independently **without** an operating system, MAC address, or IP address.

Technical Specifications:

- **Data Throughput:** Up to 1 Gbps with auto-negotiation
- **Power Supply:** 100–240 VAC, 50/60 Hz, with dual redundant internal hot-swappable AC power supplies
- **Form Factor:** 1U rack-mountable appliance
- **Mounting Options:** 19-inch 1U rack mount
- **Operating Temperature:** 32°F to 113°F
- **Storage Temperature:** -40 °F to 185 °F
- **Relative Humidity:** 5% ~ 95% | 55°C, non-condensing
- Processor: Intel i3 or better
- **RAM –** Customizable as per requirement
- **Local Storage –** Customizable as per requirement
- **Network Interfaces:** 1 x Data Port, 1 x Management Port per side
- **Relative Humidity:** - 5% ~ 95% | 131°F, non-condensing
- **In-door use only**

All images are for illustrative purposes only. Product configurations, specifications, and features may be customized to meet customer-specific requirements and may vary accordingly.

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